



MANUEL S. ENVERGA UNIVERSITY FOUNDATION

COLLEGE OF COMPUTING & MULTIMEDIA STUDIES

CHED CENTER OF DEVELOPMENT IN INFORMATION TECHNOLOGY

ITWM106

Mobile Application Development 3 (cross-platform)



MANUEL S. ENVERGA UNIVERSITY FOUNDATION

COLLEGE OF COMPUTING & MULTIMEDIA STUDIES

CHED CENTER OF DEVELOPMENT IN INFORMATION TECHNOLOGY

Introduction Cross- platform Mobile Applications Development



ccms@mseuf.edu.ph



@mseufCCMS

www.mseuf.edu.ph

Activity

- Think of three (3) meaningful questions related to the topic that will be discussed in class.
- Write each question clearly on a piece of paper
- Do not write the answers to your questions.
- Your questions should encourage discussion and demonstrate understanding or curiosity about the topic.



What is Mobile Application Development?

- Mobile application development is the process of designing, building, testing, and deploying software applications that run on mobile devices such as smartphones and tablets.



MANUEL S. ENVERGA UNIVERSITY FOUNDATION

COLLEGE OF COMPUTING & MULTIMEDIA STUDIES

CHED CENTER OF DEVELOPMENT IN INFORMATION TECHNOLOGY



ccms@mseuf.edu.ph



[@mseufCCMS](https://www.facebook.com/mseufCCMS)

www.mseuf.edu.ph

Choosing Mobile Development Framework

- What factors should be considered when selecting a mobile application development framework?



MANUEL S. ENVERGA UNIVERSITY FOUNDATION

COLLEGE OF COMPUTING & MULTIMEDIA STUDIES

CHED CENTER OF DEVELOPMENT IN INFORMATION TECHNOLOGY



ccms@mseuf.edu.ph



[@mseufCCMS](https://www.facebook.com/mseufCCMS)

www.mseuf.edu.ph

Choosing Mobile Development Framework

- These are some few considerations in choosing mobile application development framework:
 - Time to Market
 - Development Cost
 - Hiring
 - Maintenance Cost
 - App Performance
 - Feature Availability
 - 3rd Party Risks

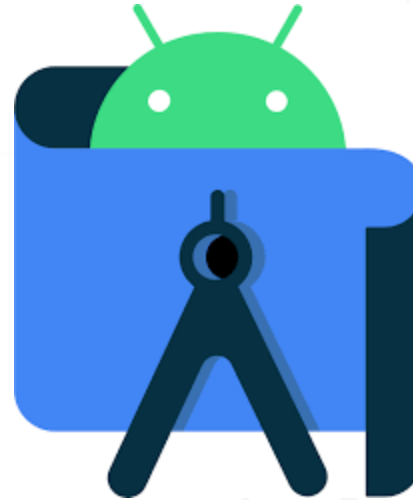
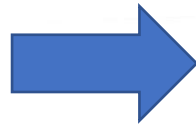


1st Party Native Applications

- Developed specifically for one operating system.
- Advantages
 - Best performance
 - Full hardware access
 - Better user experience
- Disadvantages
 - Two separate codebases
 - Higher development cost



1st Party Native Apps



1st Party Native Apps

Owner	Apple/Google
UI/UX	Native
Development Speed	Slow
Maintenance Cost	High
Performance	Very High
Programming Language	Swift/Java/Kotlin
Maturity	Very Mature
Feature Access	Complete

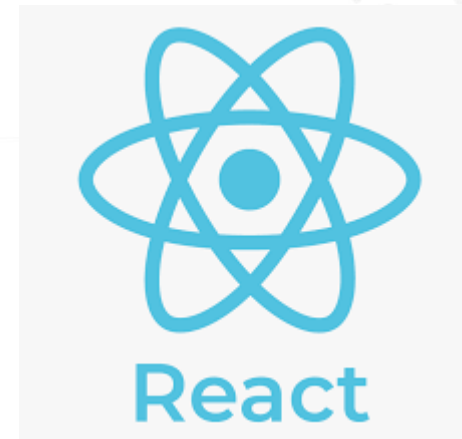


Progressive Web Apps

- A progressive web application is a type of application software delivered through the web, built using common web technologies including HTML, CSS and JavaScript.
- It is intended to work on any platform that uses a standards-compliant browser, including both desktop and mobile devices



Progressive Web Apps



Progressive Web Apps

Owner	Multiple
UI/UX	Bespoke
Development Speed	Fast
Maintenance Cost	Medium
Performance	Medium
Programming Language	Multiple
Maturity	Adolescent
Feature Access	Limited



Progressive Web Apps

- Feature Access
 - ✓ Bluetooth
 - ✓ Local Notifications
 - ✓ Touch Gestures
 - ✓ Geolocation
 - ✓ Camera
 - ✓ Device Motion
 - ✓ Offline Storage
 - ✓ File Access



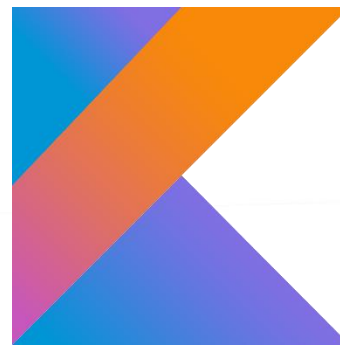
Progressive Web Apps

- Not Available

Feature	Availability
Background Geofencing	Cannot continuously monitor location when the app is closed.
Background GPS Tracking	Limited or unavailable compared to native apps.
Deep Hardware Integration	PWAs cannot fully interact with device hardware like native apps. They have limited access through web APIs.
Battery Status	The Battery Status API is no longer supported by most browsers due to privacy concerns.
File System Access	Limited compared to native apps and varies by browser.
Background Services	Cannot run continuously in the background like native apps.



Cross-Platform Native Apps



Kotlin Native



MANUEL S. ENVERGA UNIVERSITY FOUNDATION
COLLEGE OF COMPUTING & MULTIMEDIA STUDIES
CHED CENTER OF DEVELOPMENT IN INFORMATION TECHNOLOGY



ccms@mseuf.edu.ph



@mseufCCMS

www.mseuf.edu.ph

Cross-Platform Native Apps

	Xamarin	React Native	Flutter
Owner	Microsoft	Facebook	Google
UI/UX	Native	Native	Bespoke
Development Speed	Fast	Fast	Fast
Maintenance Cost	Low	Low	Medium
Performance	High	High	High
Programming Language	.Net, C#, F#	Javascript	Dart
Maturity	Mature	Mature	Young
Feature Access	Complete*	Complete*	Complete*



What is Cross-Platform App Development?

- **Cross-platform** development points to the process of creating an app that works on **several platforms**.
- This is done by using tools like **React Native**, **Xamarin**, and **Flutter**, where the apps created can be deployed on both Android and iOS.
- While cross-platform development saves time and cost, you risk sacrificing quality in the process.
- It is difficult to tailor an app that runs optimally on various platforms, and the app will need an **additional abstraction** layer when running, thus resulting in lower performance.



Advantages of Cross-Platform App Development

- **Cost Efficient** - Instead of having two teams of developers, you'll need only one to create a cross-platform app. Therefore, you'll save on the development cost.
- **Faster Development** - Only a single cycle of development is needed to create an app that runs on multiple platforms.
- **Single Code Base** - As the app is created with a single cross-platform development tool, only one code base is created.



Disadvantages of Cross-Platform App Development

- **Slower App** - The need for an additional abstraction layer and rendering process makes the cross-platform app slower than its native counterpart.





MANUEL S. ENVERGA UNIVERSITY FOUNDATION

COLLEGE OF COMPUTING & MULTIMEDIA STUDIES

CHED CENTER OF DEVELOPMENT IN INFORMATION TECHNOLOGY

ITWM106

Mobile Application Development 3 (cross-platform)